

Viability of thorium-based nuclear fuel cycle for the next generation nuclear reactor: Issues and prospects

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ABSTRACT

This paper presented a review on the past, present and the future of thorium fuel cycle. The aim of this study is to evaluate the developments in thorium fuel cycle, looking at the prospects and drawbacks on the possibility of thorium-based nuclear fuel for commercial reactors realising the increasing challenges of uranium-based nuclear fuel. The initial interest on thorium-based nuclear fuel and why it was abandoned at the early stage of nuclear technology were considered. Also, the reasons behind the present renewed interest on the viability of thorium fuel cycle as a valuable alternative to the conventional uranium-based fuel were studied. Thorium abundance, its physical, chemical and neutronic properties were evaluated in comparison to the uranium fuel cycle to determine thorium fuel sustainability for next generation nuclear industry. In this study, it was found that thorium fuel is three to four times more abundant, has higher conductivity, high melting temperature, low expansivity and more proliferation resistant compared to uranium fuel. The possible application and related challenges of thorium fuels in different reactor types and designs such as light water reactors (LWRs), high temperature gas-cooled reactors (HTGRs), heavy water reactors (HWRs), molten salt reactors (MSRs) and accelerator driven system reactors (ADSRs) were reviewed. The findings indicate that thorium fuel cycle can be used in the currently dominant LWRs designs in the nuclear industry with little technical modification, and also in other reactor types under investigation for future application especially molten salt breeder reactors, fast reactors and accelerator driven system reactors. Finally, this review made some recommendations on the short-term and long-term applications of thorium-based nuclear fuel cycle, and the issues that must be addressed before using thorium fuel for commercial reactor operations.

1. Introduction

It is a well-known fact that nuclear energy is at the centre of the world's debate politically, socially, economically and otherwise. This debate mainly is because of the increasing energy needs of the ever-growing world population, and how to manage all issues arising from the nuclear technology energy sources [1,2]. The prediction by the United States Department of Energy (US-DoE) that electricity demand will increase by 2035 is a pointer that a more reliable energy technology must evolve for commercial purposes. Since then, through relentless efforts, today many countries are meeting 20–40% of their energy needs from nuclear power using fissile uranium-235 and plutonium-239 based fuel cycles [3]. The interest in uranium-235 was due to its fissile nature compared to the Thorium-232 which is a fertile nuclear material [4,5].

However, the continuous depletion of uranium reserves, the vast production of both weapon and reactor grade plutonium and trans-uranic isotopes, issues of lack of waste repositories, and nuclear terrorism have raised serious global concerns on the continuous use of conventional nuclear reactors operating on uranium-based fuel [6,7]. Also, the withdrawal of the Yucca Mountain Repository licensing application by the US-DoE created more concerns on nuclear waste management [7,8]. Moreover, the unfortunate nuclear accidents (Three Mile Island and Chernobyl) of 1980s [9], and the US policy shift on energy investment, generally affected the advancement in nuclear industry. Despite the challenges associated with uranium-based fuel, many countries still have vested interest in well-established uranium-based fuel reactor, as other alternative such as use of thorium based nuclear fuel technology is still not available in application level. But issues arising from the continuous use of uranium-based nuclear fuel for

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the operation of over 80% reactors in existence, have greatly renewed an interest on thorium as an efficient option [10]. The most recent Fukushima Daiichi nuclear accident in Japan has equally negatively changed peoples' perception on uranium-based fuel reactors [11–14], thereby prompting greater demand for safer alternative energy sources. These issues have reawakened the need for a viable and safer nuclear fuel alternative that would reduce to a bearable minimum the challenges posed by the uranium-based nuclear fuel [15,16].

Thorium-based nuclear fuel has been of interest alongside uranium from 1950 to 1970. In fact, there are experimental and research reactors that operated on thorium-based nuclear fuel in the 1960s [17]. The 21st century has seen an increase in thorium related researches, but many has been on academic related publications and reports [17,18]. Currently, there is a renewed interest in the possible use of thorium-based fuel to generate nuclear power (electricity) at a commercial scale. India has maintained its quest for a commercial thorium-based fuel nuclear power due to their large thorium reserves with little or no uranium reserves. In fact, China and USA are currently collaborating in an intensive research programme on thorium utilisation in molten salt reactor designs [19]. Indeed, Generation IV reactor project funding which has a time frame of the year 2030, has thorium-based nuclear fuel fully recommended and incorporated molten salt cores and very high temperature gas cooled thorium-based TRISO-fuel reactors [8,20]. The renewed interest could also be attributed to the new Atomic Act on Peaceful Utilisation of Nuclear Energy and Ionising Radiation, which was published as Act No. 263/2016 Coll., replacing the previous Act No. 18/1997 Coll. This Act became effective from January 2017, and it is in fulfilment of IAEA and Euratom directives on nuclear energy utilisation [21].

This review is to evaluate the developments in thorium fuel cycle, looking at the prospects and drawbacks on the possibility of thorium-based nuclear fuel for commercial reactors keeping in mind the increasing challenges of uranium-based nuclear fuel. It also evaluated the type of reactor designs and configurations that would be suitable for this application.

2. Nuclear fuel resources

To meet the ever-increasing world population energy demand in a more environmentally friendly approach, attention has been shifted to the potential of nuclear energy that may play a greater role in the future energy mix. The sustainability of this energy mix is dependent on the availability of the sources of the nuclear energy fuel [22]. Generally, the survey and exploration of nuclear fuel resources to determine their world reserves are categorised into; conventional, unconventional and secondary resources. This categorisation is based on the naturally known and predictable resources of such potential fuel. The availability of thorium resources in large quantities is one of the strong factors encouraging and driving investors both government and private companies to committing their resources into the research [22].

Today, there is unarguably a reliable information on uranium nuclear fuel resources at various grades and levels based on the material and technological advancements made in its application in nuclear power both in military and civilian reactors. About 80% of reactors in operation worldwide are uranium-based PWRs and have been in operation for around a century. Many of the known reserves must have been depleted after a very long period of its application in reactor operation. The uranium reserves as shown in Table 1 is a clear reflection of small quantity of uranium available for current reactor operations. This shows the estimates of different categories of uranium resources by the year 2000. From the table, the undiscovered resources are over 75% of the total conventional resources while known conventional resources is less than 15%. This implies that the known will finish in no distance time. According to Red Book data 2010, the conventional uranium resources will last for more 75 years counting from 2001 at demand rate of 2,000tU per year [10,22,23]. The story is quite different for thorium,

Table 1
World uranium resources categories [10].

Resource type	Estimate (1000 tU)
Known conventional resources	
Reasonably assured resources (RAR)	2850
Estimated additional resources I (EAR-I)	1080
Undiscovered conventional resources	
Estimated additional resources II (EAR-II)	2330
Speculative resources (SR)	9940
Secondary resources	
Commercial inventories	220
Surplus military inventories	250
Re-enrichment	440
Sub-total	17,110
Undiscovered resources	
In phosphates	22,000
In seawater	4,000,000
Grand total	4,039,110

as information is sparsely available, due to lack of dedicated and continuous interest after series of events that made uranium a popular option at the period 1940s–1980s [24].

2.1. Thorium availability

It has been widely reported that thorium is three to four times more abundant [4,25], in nature than the conventional uranium, based on its average concentration in the earth's crust. Thorium exists majorly as monazite ore in the earth's crust mixed with other rare earth elements (REEs). Monazite sand is found in large quantities in India, China, Brazil, United States of America, and as well as Australia. Today, thorium is entirely produced as a by-product in the extraction of rare earth elements or from the production of titanium or zirconium bearing minerals from monazite sand. This is because of low industrial demand of thorium, since it is yet to be used for commercial production of energy in nuclear reactors. Therefore, little exploration of thorium has been made, which clarifies why there is uncertainty on its total world reserves [18,26].

Table 2 provides information on the known thorium reserves for almost all the potential countries. Low technological advancement, little or no demand for the nuclide in large quantities, and its non-application in commercial reactors, made any investment on its survey and exploration unattractive many years ago. As interest in thorium-based fuel reactor has been renewed, there is hope that more thorium deposit sites shall be discovered and quantified, raising the resource reserves. Interestingly, the reported large availability and relative ease of extraction of thorium is a good incentive for its future large-scale production as a nuclear fuel [27].

3. Thorium as a nuclear fuel

The first study on the viability of thorium as a nuclear fuel started in USA almost the same time that of uranium and plutonium under Manhattan project [5]. Thorium (^{232}Th) as a fertile isotope undergoes nuclear transmutation to produce a fissile ^{233}U nucleus. By thermal neutron capture, thorium fuelled reactor produces ^{233}U by the following likely reactions in Eqs. (1) and (2). The fissile ^{233}U either fission during nuclear reaction or may be removed chemically from spent fuel, and then forms a new nuclear fuel. It is on record that 90% of bred ^{233}U fissions while only 10% goes into spent fuel, which represents a higher fuel burn-up compared to uranium [5,6].

However, to start up a thorium fuel cycle by fissioning ^{233}U , it entails providing enough fissile material in the reactor core. This could be done by the high enrichment of thorium with ^{235}U or ^{239}Pu . The enrichment of thorium for more than 19.9%, and high plutonium are not desirable from the proliferation and waste point of view, since this

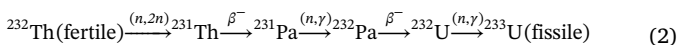
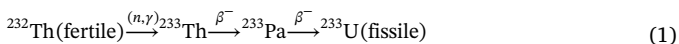
Table 2
The identified conventional thorium resources [28].

Region	Country	Total thorium resources (tTh in situ)
Europe	Turkey*	374,000
	Norway	87,000
	Greenland (Denmark)	86,000–93,000
	Finland*	60,000
	Russian Federation	55,000
	Sweden	50,000
	France	1000
	Total	713,000–720,000
Americas	United States	595,000
	Brazil	632,000
	Venezuela*	300,000
	Canada	172,000
	Peru	20,000
	Uruguay*	3000
	Argentina	1300
	Total	1,723,300
Africa	Egypt*	380,000
	South Africa	148,000
	Morocco*	30,000
	Nigeria*	29,000
	Madagascar*	22,000
	Angola*	10,000
	Mozambique	10,000
	Malawi*	9000
	Kenya	8000
	DR Congo*	2500
	Others*	1000
	Total	649,500
Asia	CIS* (excluding Russian Federation)	1,500,000
	-Including Kazakhstan, estimated	> 50,000
	-Including Russian Federal Asian part estimated	> 100,000
	Uzbekistan	5000–10,000
	India	846,500
	China estimated	> 100,000 (including 9000 ^a Tapei)
	Iran, the Islamic Republic*	30,000
	Malaysia	18,000
	Thailand*, estimated	10,000
	Vietnam*, estimated	5000–10,000
	Korea Republic*	6000
	Sri Lanka*, estimated	4000
	Australia	595,000
	Total	> 2,647,500–2,684,500
	World total	6,355,300–6,372,300

CIS, Commonwealth of Independent States.

* Data not updated in 2013.

opposes the envisaged advantage of lower concentration of transuranic elements as major contributors to long-term radiotoxicity in spent fuel and global nuclear terrorism. By this, addition of thorium in open fuel cycle is often questioned because of the present reactor design configurations [29,30].



From Eq. (2), ^{232}U is more of a contaminant to ^{233}U in the reaction chain of thorium fuel cycle, and therefore, makes it difficult to separate fissile ^{233}U because of its radiation hazard [8]. This contaminant decays to emit high energy gamma radiation isotopes (for example Tl-208) that will require the use of remote handling equipment to avoid excessive radiation exposure. Nevertheless, the contaminant is not a big threat provided it remains in the reactor, as it will be burnt during energy production process [31]. These reactions possibilities are shown in Fig. 1. Interestingly, Ashley et al. [31,32] stated that uranium-233 must not be contaminated with uranium-232, because there exist a small-scale chemical technique allowing extraction of pure of protactinium

which decays directly to a pure uranium-233. Hence, Thorium-232 is greatly a proliferation resistant nuclear material [8].

Table 3 presented the neutronic properties of fissile and fertile materials. The thermal neutron capture cross-section for ^{232}Th is more than 2 times greater than that of ^{238}U , but resonance capture cross-section is more than 3 times higher in ^{238}U than ^{232}Th . This high resonance capture cross-section is an energy factor that made ^{238}U good for fast reactor breeder material regardless of the inherent radiotoxicity and proliferation issues. ^{232}Th is a very good breeder candidate for thermal reactors, thereby making it the best fertile nuclear material applicable in the current thermal reactors that are in operation. ^{233}U depends less on neutron energy, as its breeding ratio is small (> 1) at thermal region and relatively large (> 2) at epithermal and fast regions. Further, the resonance fission cross-section for ^{233}U is almost three times more than that of ^{239}Pu . The striking advantage of ^{233}U over ^{235}U and ^{239}Pu in a breeder reactor is on its high capture-to-fission ratio, $\alpha = (\sigma_f/\sigma_f)$ compared to ^{239}Pu . The implication is that if α is lower, the chance of fission reaction occurring per neutron absorption increases instead of capture reaction [35]. Hence, this cause of high neutron yield per absorption, $\eta = \nu \frac{\sigma_f}{(\sigma_f + \sigma_a)}$ in a thermal spectrum up to 100 keV for ^{233}U compared to others. This high neutron yield per absorption made ^{233}U a good neutron economy in a reactor. These advantages made ^{232}Th an important nuclear material for potential application in nuclear industry [27,36,37].

Where:

α = neutron capture-to-fission ratio

σ_f = neutron capture cross-section

σ_f = fission cross-section

$\sigma_a = \sigma_f + \sigma_f$ = absorption cross-section of fuel

ν = number of neutrons produced per fission reaction of fuel

η = number of neutrons produced per neutron absorption by fuel

3.1. Thorium utilisation in reactors

The idea of using thorium for nuclear fuel came following its proposal by Eugene Wigner in 1944 to the New Piles Committee. The committee came up with a recommendation that more research should be done on possible thorium nuclear fuel application. Following Manhattan project, experiments were conducted and discovered that thorium is not fissile rather fertile, and therefore can breed fissile ^{233}U by neutron capture reaction and subsequent beta decays. It was hoped that ^{233}U had the potential for nuclear weapons [5], which was the major interest of United States at that very early stage of nuclear technology. The non-fissile nature of thorium and non-weapon potential of ^{233}U decreased the interest of its possible immediate application especially in the United States who had the interest of nuclear weapon. However, interest in thorium option was never completely discarded because of presumed limited uranium sources, as the nuclear technology was not matured at that time. This was done in parallel to their demand for nuclear weapon proliferation. In view of this, many countries such as Germany, UK, Holland, France etc. led by the USA developed prototype thorium-based nuclear reactors [38]. Table 4 presented the thorium-based fuel cycle experiment and test type reactors developed and operated by different countries. This shows that USA made the highest contributions and is leading in the research. The highest power generated is 300 MW from the German THTR-300 and Fort St-Varen HTR in USA.

Interestingly, these researches have revealed that thorium-based fuel has the potential of being used in most reactor types using the conventional uranium fuel in its operation without major technical modification. Radkowsky proposed a seed-blanket reactor design that would operate on thorium-based nuclear fuel which hopefully would replace Radkowsky uranium-based designs. This led to the establishment of Radkowsky Thorium Power Corporation in the USA in 1992

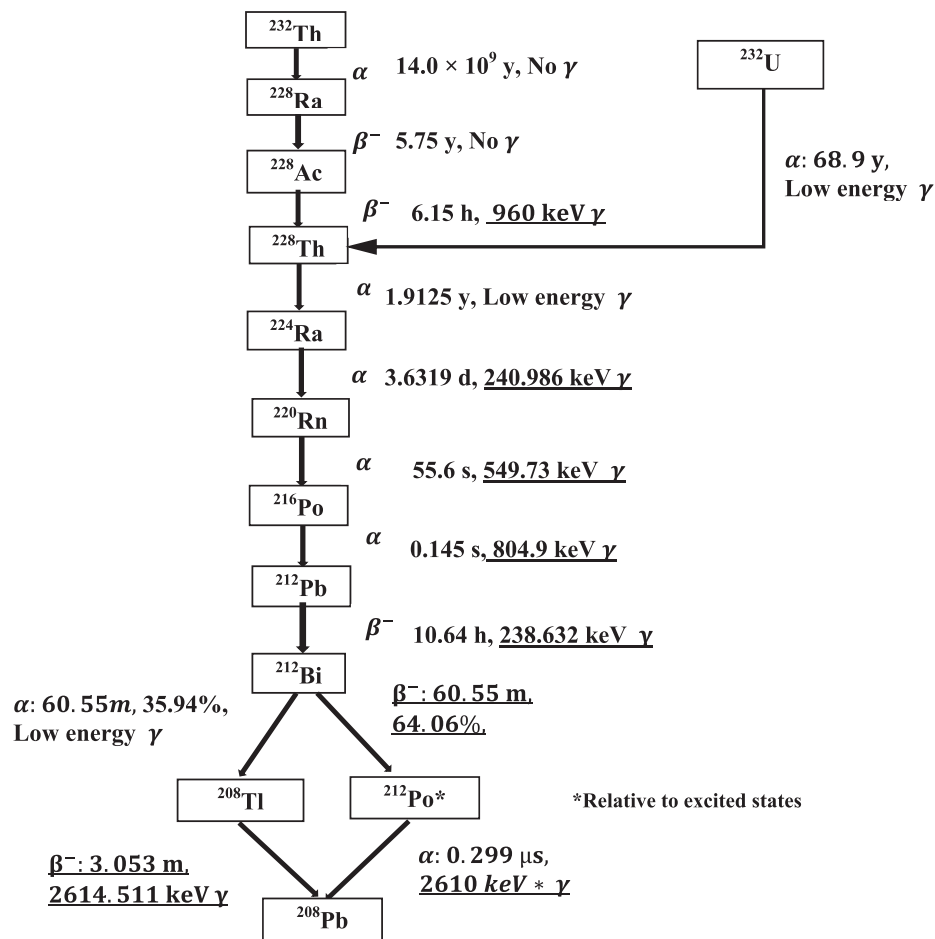


Fig. 1. The alpha and beta decays, and the products of Th-232 and U-232 decay chain [33,34].

Table 3

Nuclear data for fissile and fertile materials [27].

	Average over thermal spectrum			Resonance integrals			Average over fission spectrum		
	σ_c	σ_f	ν	σ_c	σ_f	ν	σ_c	σ_f	ν
Fissile materials									
^{233}U	42.20	468.2	2.495	134.16	751.71	2.498	0.063	1.84	2.596
^{235}U	86.70	504.81	2.433	131.97	271.53	2.438	0.095	1.219	2.583
^{239}Pu	274.32	699.34	2.882	184.06	289.36	2.876	0.065	1.8	3.091
^{241}Pu	334.11	936.65	2.946	169.13	570.66	2.933	0.226	1.62	3.151
Fertile materials									
^{232}Th	6.533	–	–	84.97	–	–	0.102	0.071	2.093
^{238}U	2.414	1.05×10^{-5}	2.489	277.7	2.16×10^{-3}	2.49	0.07	0.3	2.598
^{240}Pu	262.65	6.13×10^{-2}	2.784	8448.7	3.74	2.785	0.095	1.349	3.013

[39,40]. This corporation worked in close collaboration with the Russian Research Centre, Kurchatov Institute, David Ben Gurion University in Israel, Brookhaven National Laboratory and Massachusetts Institute of Technology (MIT) in USA. This collaboration was to jointly understand the Russian reactor such as VVER-440 and VVER-1000 operating on Radkowsky thorium-based nuclear fuel. It also had interest in the Russian proposed 2005, low water test reactor [41]. All these research collaborations were aimed at producing thorium-based fuel reactors that will operated at commercial and larger scale, thereby reducing the stockpiled plutonium and minimise difficulties in waste management and proliferation. One of the hallmarks of thorium-based nuclear reactor is on its ability to generate cleaner and environmentally friendly energy, which achieves the sustainable development goal of energy sources that solve the present needs without short-changing the energy

needs of future generation [42]. The added advantage of nuclear energy is that the green-house gas emission of nuclear power generally is comparable to that of solar and wind energy sources [39].

3.2. Thorium fuel types and properties

A comprehensive information on different types of thorium-based nuclear fuels used in different types of reactors is presented in this section. A comparative analysis of the properties of the fuels is also been investigated. Thorium-based nuclear fuels have fared very well in so many reactors as shown in Table 4 due to its basic physical and chemical properties. Studies have shown that natural thorium (^{232}Th) alone cannot be used as nuclear fuel as it is a fertile nuclide just like U-238 that requires a fissile nucleus for fertilisation to occur before fission

Table 4
Thorium utilisation in experimental and nuclear power reactor [43].

Name and location	Type of reactor	Power	Fuel	Period
CIRUS, India	MTR, thermal	40MWt	“J” rod of Th and ThO ₂	1960–2010
Indian point, USA	LBWR PWR, (pin assembly)	285MWt	Th + ²³³ U (oxide pellets)	1962–1980
Elk river, USA	BWR	24MWt	Th + ²³⁵ U + ²³⁸ U (oxide pellets)	1963–1968
MSRE, ORNL, USA	MSR	7.5MWt	U ²³³ molten fluoride	1964–1969
Peach bottom, USA	HTGR, experimental (prismatic core)	40MWt	Th + ²³⁵ U, Coated fuel particles, Oxide & Dicarbides	1966–1972
Dragon, UK OECD- Euratom	HTGR, Experimental (Pin-in-Block Design)	20MWt	Th + ²³⁵ U, Coated fuel particles, Oxide & Dicarbides	1966–1973
AVR, Germany	HTGR, Experimental (Pebble-bed reactor)	15MWt	Th + ²³⁵ U, Coated fuel particles, Oxide & Dicarbides	1967–1988
Lingen, Germany	BWR (Irradiation testing)	60MWt	(Th, Pu)O ₂ pellets	1973
SUSPOP/KSTR KEMA, Netherland	Aqueous Homogenous Suspension	1MWt	Th + HEU, Oxide pellet	1974–1977
Fort St Vrain, USA	HTGR, power (Prismatic core)	330MWt	Th + ²³⁵ U, Coated fuel particles, Dicarbides	1976–1989
Shippingport, USA	LWBR PWR, (Pin Assembly)	100MWt	Th + ²³³ U (Oxide pellets)	1977–1980
THTR-300, Germany	HTGR, Experimental (Pebble-bed reactor)	300MWt	Th + ²³⁵ U, Coated fuel particles, Oxide & Dicarbides	1985–1989
FBTR, India	LMFBR, Pin Assembly	40MWt	ThO ₂ Blanket	1985-present
DHRUVA, India	MTR Thermal	100MWt	“J” rod of ThO ₂	1985-date
KAMINI, India	MTR, Thermal (for Research)	30kWt	Al + ²³³ U	1996-date
KAPS 1 & 2; KGS 1 & 2; RAPS 2, 3, & 4, India	PHWR, (Pin Assembly)	220MWt	ThO ₂ pellets for initial flux flattening	new existing
NRU & NRX, Canada	MTR (Pin Assembly)	–	Th + ²³⁵ U, Test fuel	–

takes place. Therefore, it has always been supplied or mixed with fissile sources like ²³⁵U, ²³⁹Pu or neutron sources like spallation neutrons from ADS [6].

However, thorium oxide (ThO₂) on its own has a very stable structural, thermodynamic and thermophysical properties in comparison with UO₂ and PuO₂ oxides. Thorium oxide fuel is distinguished over uranium and plutonium oxides because of its single oxidation state (+4) in compounds [27]. This means that thorium oxide has a low number of oxygen vacancies and interstitial defects, showing a very little non-stoichiometry. This indeed is reflected on its high thermal conductivity, high thermal stability, defined oxygen potential, low chemical reactivity, low matrix diffusion, and difficulty in pellet sintering [44]. Also, Th-MOX fuel material properties greatly outweighed the conventional mixed oxides (MOX) fuel [4], as discussed in Section 3.2.1. Table 5 shows the comparison of the physical properties of thorium, uranium and plutonium dioxides. From the table, it is observed that the three oxides have face-centred cubic (fcc) lattice crystal structure, and they can be heated to melt without phase transition, therefore, showing great similarities.

3.2.1. Material properties of Th-MOX fuel

Generally, the material properties of thorium mixed oxide (Th-MOX) fuel have been found most reliable and efficient over the conventional uranium and plutonium mixed oxides fuels. The introduction of conventional MOX fuel into LWRs posed many challenges more than the UOX fuel. The conventional MOX fuel in LWRs posed problems of reduction in fuel thermal conductivity, high fuel temperature, higher concentration of transuranic nuclides in spent fuel, and increased

Table 5
The physical parameters of thorium, uranium and plutonium dioxides [27].

Property	Unit	ThO ₂	UO ₂	PuO ₂
Lattice structure	–	fcc	fcc	fcc
Lattice constant, a	Å	5.597	5.4702	5.3815
Cell volume, V _c	Å ³	175.3	163.7	157.5
Molecular weight, M _w	au.	264.04	270.03	276.00
Theoretical density, ρ	g cm ^{−3}	10.00	10.96	10.46
Melting temperature, T _m	K	3651 ± 17	3120 ± 30	3017 ± 24
Standard enthalpy of formation, ΔH ^o	J mol ^{−1}	−1226.4	−1085.0	−1055.8
Standard entropy, S ^o	J mol ^{−1} K	65.23	77.03	66.13
Heat of fusion, H _f	J mol ^{−1}	90 ± 00	78 ± 00	67 ± 15

From previous studies, PuO₂ melting temperature is about 300 K, and the molecular weight of plutonium is conventionally fixed at 244 since it has no stable isotopes [45].

fission gas release. But the introduction of Th-MOX fuel in same reactor removed these challenges, and even perform better than UOX [4]. Currently, less than 10% of thorium is introduced as an additive in conventional UOX fuel for use in LWR. The benefit of this has been in increased neutron absorption over uranium, which helps to reduce the use of reactor poison in flattening reactor power distribution, and therefore, increase thermal conductivity, reduce fuel cracking, and improve reactor safety parameters [46,47].

Thorium mixed oxide fuel properties change as a function of the composition of its components, which affects the dissolution rate of the fuel depending on the percentage of thorium present [48]. (Th, U)O₂ is used today in conventional LWRs for power peaking control and reduction of the need for reactor poisons [4,47]. Fig. 2 shows the thermal conductivity of pure thorium oxide, uranium oxide, and thorium mixed oxides fuel studied at 20% of UO₂, and 14% and 7.9% composition of PuO₂ for (Th, U)O₂ and (Th, Pu)O₂ fuels, respectively. It can be seen that ThO₂ fuel has the highest and distinct thermal conductivity at all the temperature range except at low temperature followed by 7.9% PuO₂ of (Th, Pu)O₂ at low temperature. The 7.9%PuO₂ of (Th, Pu)O₂ overlapped with that of 20%UO₂ of (Th, U)O₂ at temperature ≥ 1200 °C while 14%PuO₂ of (Th, Pu)O₂ has the least thermal conductivity which overlapped with UO₂ at temperature ≥ 600 °C up to 1600 °C. From the results, one can conclude that the presence of thorium dioxide in the fuel mixtures impacted positively on their thermal conductivities. Therefore, it is necessary that more researches should be conducted by varying the percentage composition of the dioxides to help determine that which would be suitable for reactor fuel

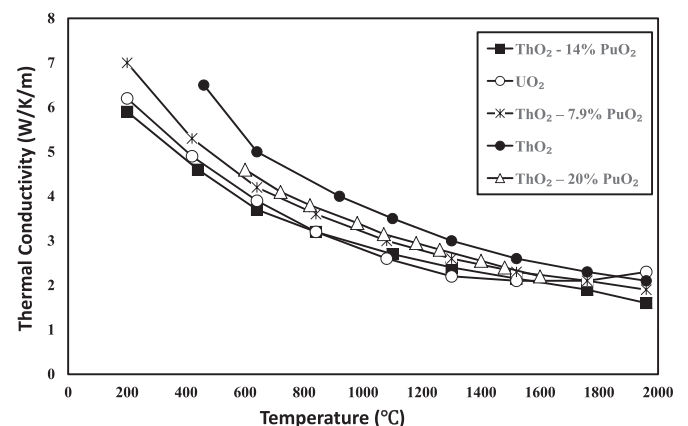


Fig. 2. The thermal conductivity of (Th, U)O₂, (Th, Pu)O₂, and pure UO₂ and ThO₂ [4].

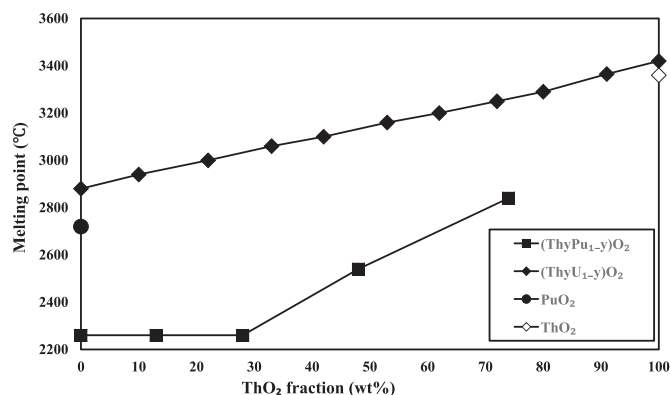


Fig. 3. The melting temperature of (Th, U)O₂, (Th, Pu)O₂, and pure PuO₂ and ThO₂ [4].

operation [4,49]. Fig. 3 shows that the melting temperature of (Th, U)O₂ is higher compared to the melting temperature of both UO₂ and (Th, Pu)O₂, thereby increasing the melting temperature margin of this fuel in the reactor core. Indeed, the effects of the higher thermal conductivity of these fuels are that it lowers the fuel temperature in the reactor core, increasing the melting temperature of the fuel and gives a lower fission gas release (FGR) [46]. These benefits in Th-MOX fuels are due to the stable physical and chemical properties exhibited by thorium dioxide in the fuel. By this, thorium gives better radiation resistance resulting to an improved fission gas release property of the fuel as less FGR will be recorded. The FGR of the fuel must be limited because it decreases the attainable fuel discharge burn-up [50].

3.3. Thorium-based fuel reactors and design configurations

From Section 3.1, it is known that thorium can be used to generate nuclear power. This is achievable using fertile thorium or thorium oxide fuel and different fissile fuel or its oxide combinations depending on the reactor type and design configuration [51]. Thorium fuel has shown good course and potential in high temperature gas cooled reactors (HTGRs), light water reactors (LWRs), heavy water reactors (HWRs), molten salt reactors (MSRs), and accelerator driven system hybrid reactors (ADSHRs). Apart from the issue of start-up fissile radioactive neutron source, thorium-based nuclear fuel has a good performance in reactors. The thorium-based fuel greatly minimises the issue of weapon grade materials in spent fuel, radiotoxicity and waste management (due to the presence of long-lived minor actinides), nuclear material proliferation, and global terrorism. However, several thorium-based fuel types have been used in different reactor types to investigate its performance and the possible solution to the issue of start-up radioactive fissile source needed to fertilise thorium. This is one of the major drawbacks to thorium fuel application in commercial scale in a thermal nuclear reactor. In response to this, Professor Carlo Rubbia at Centre for European Nuclear Research (CERN) in 1996, proposed the concept of Energy Amplifier or Accelerator Driven System (ADS) [52]. This process is to generate neutrons through spallation reaction without using the radioactive fissile nuclides. This is hoped to resolve permanently the issue of initial neutron sources for thorium fertilisation in reactors, the contentious issue of long-lived radioactive actinides, nuclear waste management, and proliferation risk. Although, this system has not been fully developed, serious research is ongoing at CERN and various research centres, and it is progressing accordingly. As interesting as this could be, the major concerns of nuclear technology countries is on economic loss, since the energy amplifier system requires a complete new nuclear technology. Hence, the reason for more attention dedicated to other forms of thorium-based nuclear fuel hoped to replace the conventional fuel with considerable technical modification to the reactor. These types of thorium-based reactors are preferred to the fertile

²³⁸U because they do not generate a good quantity of long-lived actinides and weapon grade materials, therefore, easing waste management challenges and proliferation treat [18].

3.3.1. High temperature gas cooled reactor

In 1947, Oak Ridge National Laboratories in United States designed a research reactor that used thorium-based fuel in its operation. This mission was actualised in Peach Bottom, PA 40-MW reactor that produced electricity from 1966 to 1974. The Peach Bottom was later upgraded to a 300 MW version operated at Fort St Vrain from 1976 to 1989. The reactors were operated with oxides and dicarbides of thorium and uranium-235, which were arrayed in prismatic graphite cores, and helium cooled [6,38]. Germany introduced thorium high temperature reactor (THTR) known as THTR-300. The reactor was a pebble-bed core with TRISO (triple-coated isotropic) thorium kernels enriched with uranium-235 embedded in graphite matrix. The kernels were stacked in a baseball spherical pebbles with graphite outer layer. THTR-300 operated from 1983 to 1989 producing 300 MW power against the initial installation output of 15 MW. Indeed, more progress was made in 1999, when Japanese, thorium based TRISO fuel 30 MW high temperature test reactor became operational. This reactor contained prismatic core with enough outlets to separate hydrogen from water. China built and completed a pebble bed prototype design of HTR-10 with output power of 10 MW. With the success in HTR-10, its two full-scale Chinese 250 MW designs were to be commissioned in 2013 [8].

Furthermore, America designed a gas cooled high temperature teaching and test reactor with commissioning date of 2015. The core was to be either pebble-bed or prismatic block. Based on this, the United States led Generation IV International Forum (GIF) [53], tipped very high temperature reactor (VHTR) for hydrogen production, coal gasification, and desalination purposes alongside electricity production. This VHTR reactor is a TRISO fuel 600 MW gas turbine billed for commissioning in 2030. It is worthy to note that high temperature gas cooled reactors (HTGR) are among the Generation IV reactors that is being researched whose focus is on reactor safety and sustainability. Therefore, HTGRs have the capacity to minimise (not eliminate) the major issues of concern with uranium or plutonium-based fuel as earlier stated. HGTRs have the capacity of using thorium mixed oxide fuels (Th, U)O₂, (Th, Pu)O₂ and (U, Pu)O₂ hence, they are pursued for the near future application [7,8].

3.3.2. Water-cooled reactors

3.3.2.1. Light water reactors. Light water reactors account for over 80% of pressurised light water reactors (PWRs) in the world operating in once-through cycle system [54]. It is the main reactor type considered of what happens to the heavy investment made in it, since the time of thorium fuel cycle proposal. Previous studies have shown that thorium fuel cycle can replace uranium fuel cycle currently used in all PWRs without major technical modification [5]. Thorium fuel introduction in boiling water reactors (BWRs) also proved successful. This was the fuel used in the Elk-river reactors in Minnesota, United States. The use of thorium fuel was also tested in German's 60MWe BWR reactors. This showed the flexibility of BWR reactor design by its success in thorium fuel optimisation. Several research reactors such as Shippingport and Indian point have shown that thorium fuel cycle is viable in PWRs. Radkowsky reactor concept (that is seed-blanket) offered a great option for LWRs. This concept has been used successful in Russia's water-water energetic reactors, VVER-40, VVER-1000 and VVER-1200 [55,56]. According to Galahom [6], replacing ²³⁸U with ²³²Th will need higher enrichment with ²³⁵U as to achieve the same burn-up. It is stated that the Doppler coefficient of ²³²Th/²³⁵U fuel assembly is more negative than other proposed fuels, and that the fission rate in the case of ²³²Th/²³⁵U and ²³²Th/²³³U is also larger compared to other proposed fuel assemblies. Thorium-plutonium oxide fuel for LWRs are undergoing serious study in the Halden Research Reactor in Norway

since 2013 [57]. Interestingly, thorium fuel rod bundles have been developed by Radkowsky, and a number of these rod bundles have been found feasible for application in LWRs. The simple approach for thorium utilisation in solid core thorium reactors is to replace uranium fuel rods with thorium fuel rods in a conventional light water design configuration [58]. Indeed, the hope of using thorium fuels in PWRs boosted the chance of building commercial light water reactors using thorium fuels in the nearest future.

3.3.2.2. Heavy water reactors. Studies have shown that thorium and uranium fuels have common design properties in reactors, and many researches on the use of thorium fuel have justified this thorium and uranium fuel similar properties concept. Canada as a country mostly interested in heavy water reactors has over 50 years of research experience on thorium fuels use in heavy water reactors (HWRs). It has conducted over 25 tests in three research reactors and a pre-commercial reactor in CANDU-type reactors. India, that has about 25% of thorium reserves in the world, has maintained the use of ThO_2 pellets in pressurised heavy water reactor (PHWRs) for neutron flux control (flattening) of reactor core after start-up. It has also tested its low enriched uranium advanced heavy water reactor (AHWR-300). AHWR-300 is a 300 MW vertical, pressurised tube, using boiling light water as coolant, and heavy water as moderator. This reactor uses thorium oxide fuel enriched with 19.75% uranium oxide. Although, the enrichment is regarded as low, but it is very high compared to the current recommendation of 3–5% uranium enrichment. On the average, 39% of its power is from thorium. India has a target of year, 2050 to achieve her power requirement using thorium-based fuel reactors [8,28,37]. Some of the great features and advantages of CANDU-type reactor using either thorium or uranium oxide or mixed oxides of uranium and thorium are:

1. CANDU-type uses automated online refuelling technique while in operation without shutting down the reactor due to its horizontal fuelling design. The robotic machine pushes in the new fuel from one end and removes the spent fuel from the other end at the pressure tube.
2. As it uses heavy water as both coolant and moderator, neutron losses are fewer since the heavy water absorbs less neutron thereby ensuring neutron efficiency and economy in the reactor.
3. The pressure tube in CANDU instead of pressure vessel as in PWRs allows neutrons flow freely between tubes and fuel bundles.
4. The convectional CANDU reactor uses uranium-based fuel with the inherent issues associated with it such as the presence of 98.7% ^{238}U , 0.8% fission product, 0.4% plutonium and traces of long-lived minor actinides in the spent fuel. But thorium-based fuel will have high burnup and generate little or no ^{238}U in the spent fuel. The fission products generated will be higher but can easily be removed and stored in a geological repository.

The major drawbacks in CANDU-type reactors generally whether uranium or thorium-based fuel are on the high cost of heavy water and extremely large size of the reactor.

3.3.3. Molten salt reactors

In a bid to find lasting solution to the issues of stockpiled weapon-grade plutonium, long-lived minor actinides and nuclear waste associated with uranium-based fuel, a lot of researches were conducted. The Oak Ridge National Laboratory (ORNL) in the United States came up with molten salt reactors (MSRs) design configuration to solve and/or reduce such issues. MSRs have thorium breeder fuel dissolved in molten salt which serves as the primary coolant, with graphite moderator [59,60].

The design was to have online dispensation of fission products, and to burn the contentious minor actinides. The interesting thing about the molten salt as coolant is that it remains liquid at 500 °C to about

1400 °C without pressurisation unlike the PWRs whose operating temperature is about 315 °C at 150 atmospheric pressure. In 1954, ORNL operated a molten salt test reactor known as aircraft reactor experiment (ARE) at 2.5MWt power, which was intended for use as an engine in a nuclear-powered aircraft [61]. The aircraft reactor experiment had molten fluoride salt ($\text{NaF-ZrF}_4\text{-UF}_4$ in the ratio of 53-41-6 mol%) as its fuel with beryllium oxide (BeO) as moderator and helium as a secondary coolant, which operated for only few days. ORNL operated another one called Pratt and Whitney Aircraft Reactor No. 1 (PWAR-1) at zero power with molten fluoride salt as fuel and primary coolant. The centre later mounted an engineering test reactor called molten salt reactor experiment (MSRE) that operated at 7.4MWt. It used $\text{LiF-BeF}_2\text{-ZrF}_4\text{-UF}_4$ (65-29-5-1 mol%) as fuel, graphite core, and fertile blanket of lithium-beryllium fluoride as secondary coolant. The fuel was later changed to ^{235}U at first and then to ^{233}U which is bred from thorium.

Following the success achieved by using ^{233}U , ORNL designed 1000MWe molten salt breeder reactor (MSBR) that used $\text{LiF-BeF}_2\text{-ThF}_4\text{-UF}_4$ (72-16-12-0.4) as fuel with graphite moderator. This offered a better hope on the potential of molten salt reactor as it has a breeding ratio (BR) of 1.07. Although BR for MSBR is less than that for fast reactor, but its doubling time was almost the same [62]. These successes made some nuclear agencies in United Kingdom, Russia, and Australia to carry out theoretical and experimental research on MSR. For instance, the Flibe Energy Co. in United States had proposal for a 20–50 MW small design modular liquid fluoride thorium reactor using Li-Be eutectic salt mixtures for power in military bases. Recently, China and Japan have developed thorium fuel programme using molten salt technology. This gave birth to the Japanese 200 MW Fuji thermal breeder molten salt reactor that is fully supported by United States and Russia. MSR is very important as it is inherently safe, chemically inert, and operates under low pressure to prevent explosion and avoid release of toxic wastes. Chinese molten salt reactor is considered the largest using the technology in the world. The new Generation IV International Forum is considering MSBR and MSFR in her research and has suggested that MSRs demonstration reactors could start after 2025, which is the proposed end of the viability research and developmental period [53]. The schematic design concept of molten salt breeder reactor (MSBR) is shown in Fig. 4. MSBR is one of the Generation IV nuclear reactors with the following advantages:

1. Thorium-based fuel is its typical reactor fuel.
2. It has good reactor safety measures, and a strong negative Doppler reactivity coefficient, which means that the rate of nuclear fission decreases once there is an increase in the reactor core temperature.
3. The contentious long-lived actinides will be burned during its operation, thereby minimising the difficulty earlier encountered mostly in LWRs nuclear waste management.
4. It is proliferation resistant since the weapon-grade material will be burnt in the reactor.
5. The nuclear waste is mainly short-lived fission products, which can easily be handled in geological repository containment, unlike the LWRs and HWRs that generate long-lived actinides.
6. In a case of emergency, the fuel is drained to emergency dump tank for safety as shown in Fig. 4.

3.3.4. Accelerator driven system reactors

The concept of accelerator driven system reactor (ADSR) was proposed by Professor Carlo Rubbia in 1996 at CERN [8,64]. The driving force of this reactor concept is on its ability to utilise a variety of nuclear fuels based on uranium, plutonium or thorium, and the fact that its thorium nuclear fuel does not require fertilisation by fissile materials. The accelerator driven system would be coupled with the reactor to generate neutrons by spallation reaction. The particle accelerator delivers proton beams of about 12 MW power and 1 GeV kinetic energy, which then generates neutrons by spallation reactions in a heavy metal target. If this technology eventually becomes operational, it offers

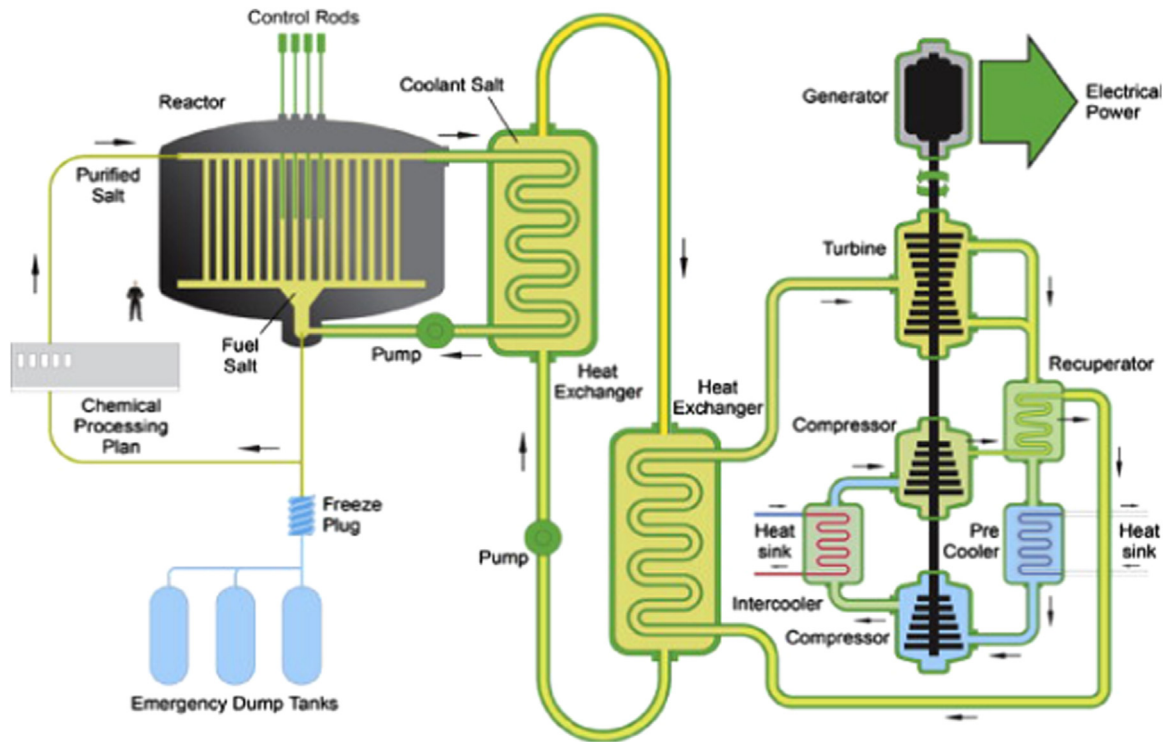


Fig. 4. The schematic diagram of the Generation IV reactor design for MSBR [53,63].

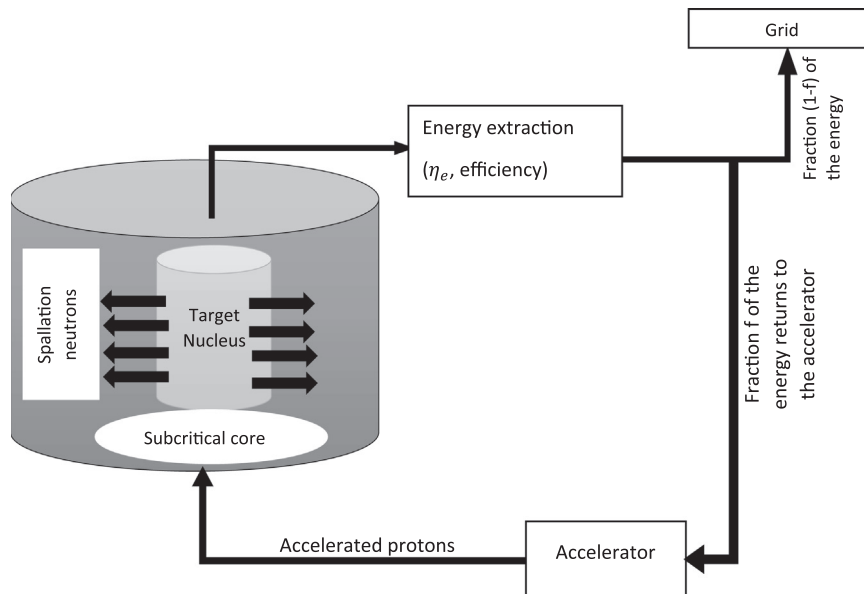


Fig. 5. The block concept of accelerator driven system reactor (ADSR) [65].

higher safety parameter, sustainability, and economic competitiveness as it would greatly minimise waste management issues surrounding today's reactors. The outstanding safety benefit of ADSR is because the reactor operates at subcritical level, therefore, fission chain reactions will not be sustained unless there is an external neutron supply (that is by ADS). It has the potential to accommodate different forms of nuclear fuel such as MOX fuel. Fig. 5 shows the block concept design configuration of an ADSR with its three main components (the accelerator, spallation target and the subcritical core) while Fig. 6 presented the core components of the reactor [65].

During shutdown operation, the reactor can be conveniently

shutdown by cutting of the proton beam source thereby stopping fission reaction leaving only the decay heat to be controlled with control rods. Currently, no ADSR is in operation but serious and intensive research is going on in many research facilities worldwide such as Oak Ridge National Laboratory, Kyoto University Reactor Institute, Japan, Indian Atomic Energy Commission, Belgian nuclear research centre and UK-Swiss proposal for 600 MW accelerator driven thorium reactor in UK. Although, ADSs still have a long-distance hope of operation, it would be the most reliable as its design configuration is more politically acceptable [8,65].

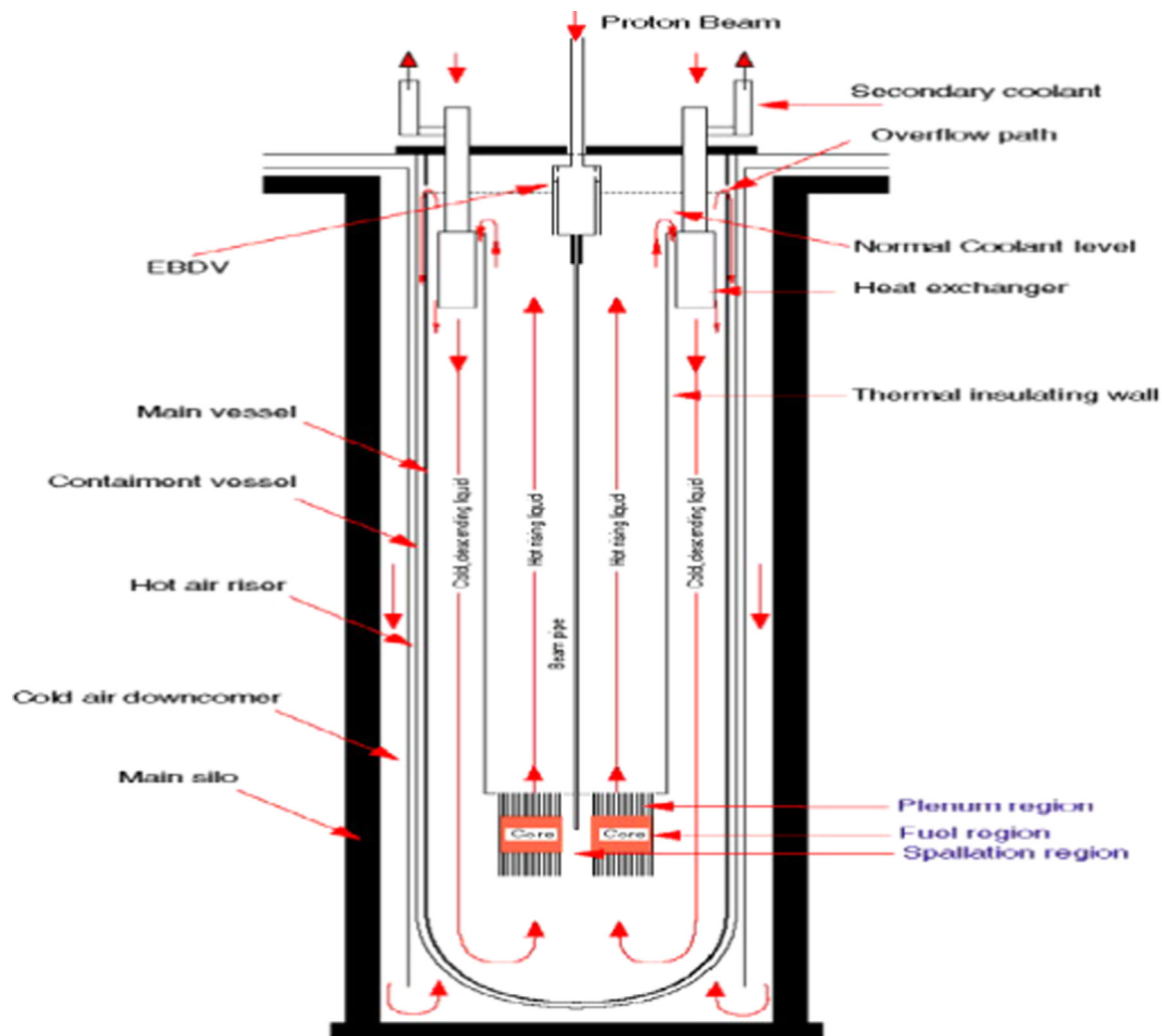


Fig. 6. General core layout of the accelerator driven system in a reactor [8].

4. Fabrication of nuclear fuel

Nuclear fuel fabrication is simply a reactor fuel manufacturing operation process, in which the key processes for both fresh and recycle fuel fabrication are the same. Fuel fabrication processes determines and add to the cost of building a nuclear reactor, which is among the factors considered by any investor. These reactor fuels are in two basic forms namely; oxide fuel pellets that are contained in metal tubes, which are later arranged in mechanical assembly for LWRs, HWRs, LWBRs and FBRs, and microspheres coated with layers of ceramic coatings contained in graphite assembly for HTGRs [66]. The processes for the fabrication of thorium-based fuel are relatively well developed for use in many reactor types [27], as fuel manufacturing process depends on its end use (reactor type). This was achieved based on the success in the fabrication of conventional MOX fuel, which is then applied in thorium-based fuel since it is mostly and efficiently used as MOX fuel. To manufacture reactor fuel, the fuel desired microstructure (morphology), chemical composition, and the stoichiometry must be generally assured and qualified [4]. Also, care must be taken to ensure that the added fissile components of the fuel are properly mixed on a microscopic level and distributed evenly to achieve uniformity [38].

Generally, reactor fuels are fabricated through three major routes which have been applied without hitch to manufacture thorium-based oxide fuel. These routes are: conventional binder less powder pressing, spheroidisation of powder-plasticiser mixtures, and the sol-gel process. The last two routes have produced fuel microspheres with diameters in

the range of 50–1000 μm , which can be pressed into pellets, and used directly in Sphere-Pac/Vipac arrangements or coated with carbon and silicon carbide layers to produce TRISO fuel for HTRs [27]. These methods have been extensively developed and used by India, USA, Germany, Canada, France etc. based on their in-depth experience in thorium fuel technology [46,67,68].

4.1. Powder metallurgical methods

The powder metallurgical method of fuel fabrication has proved successful in UOX and conventional MOX fuels [4]. This fuel manufacturing technique has been used successfully for $(\text{Th}, \text{U})\text{O}_2$ fuel fabrication [4], which involves direct blending of UO_2 and ThO_2 powder [68], while that of $(\text{Th}, \text{Pu})\text{O}_2$ is still under investigation [4]. Specifically, $(\text{Th}, \text{U})\text{O}_2$ fuel pellet was produced at Institute of Energy Technology (IFE) as a result of fuel irradiation campaign programme conducted at Halden, and commissioned by Thor Energy Company in collaboration with IFE in Norway [69]. However, the issue of control of oxygen potential during sintering as it occurs in uranium or plutonium does not arise, as thorium oxide has single + 4 valent cation, with little non-stoichiometry. Therefore, sintering of thorium oxide can be done in both oxidising and reducing conditions (i.e. air, argon, vacuum etc.). But the physical and thermochemical stability of thorium oxide decreases its sintering ability and makes production of high density fuel more difficult [27]. The powder morphology of ThO_2 shown in Fig. 7 which is plate-like unlike UO_2 that has granular structure (Fig. 8), poses

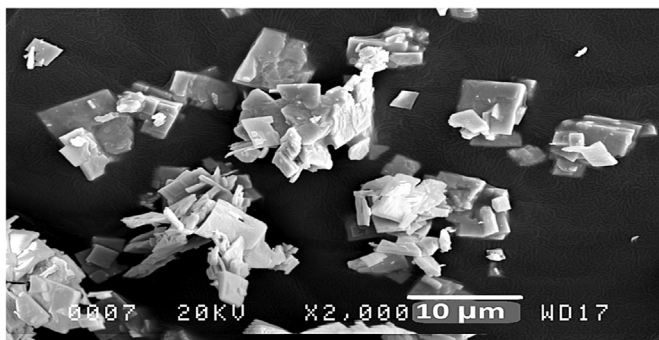


Fig. 7. The ThO_2 as-received powder which was calcined around 950°C for about 4 h. The specific surface area of this highly calcined product is less than $2\text{ m}^2\text{ g}^{-1}$ [4].

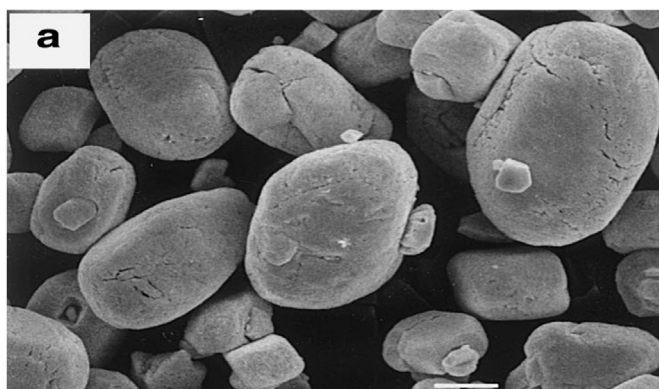
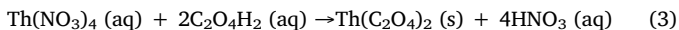


Fig. 8. The granular morphology of UO_2 [70].

fabrication challenge. To make up for the effect of ThO_2 high temperature and plate-like structure, its powder is to be subjected to a high degree of pre-processing, which if achieved, then there will be no need for high fuel sintering temperature more than that for standard uranium dioxide fuels, thereby making densities of 96% theoretical density (TD) possible to attain.

4.1.1. Powder preparation, milling and production

The thorium oxide powder is prepared by calcination process [38], using thorium oxalate, $\text{Th}(\text{C}_2\text{O}_4)_2$ [27], which is usually obtained by adding oxalic acid to a thorium nitrate solution. The chemical reaction is as shown in Eq. (3). The thorium oxide grain size obtained after calcination is around $1\text{ }\mu\text{m}$ with a platelet geometry which makes it hard to achieve high density sintered pellets [27]. The blocky nature of ThO_2 morphology requires that its powder must be milled to increase the bulk surface area [38]. Specifically, ThO_2 powder at Institute of Energy Technology (IFE) in Norway, was thoroughly prepared to achieve bulk surface area of $8\text{--}9\text{ m}^2/\text{g}$ [4].



Although, according to White et al., [71] pellets of high density value of 96% TD can be fabricated without pre-milling if the oxalate precipitation is done within a temperature range of $0\text{--}10^\circ\text{C}$ less than room temperature [27,71]. However, to co-mill ThO_2 with other oxides such as Plutonium may need dispersants to control electrostatic force which is the cause of possible particle agglomeration and dust if not controlled [4]. The control of the electrostatic force is extremely important in the production of $(\text{Th}, \text{Pu})\text{O}_2$ fuel which is still been investigated. To reduce the overall milling time, the thorium dioxide powder must be separately milled, with intermediate sieving, then followed by recycling of large platelets [4,38].

4.1.2. Pellet pressing – green pellet production

Green pellet pressing has been majorly used with double-sided press for conventional uranium oxide fuel production, basically at IFE in Norway. Here, standard dye lubricant is employed in the pressing while organic or other binders are avoided in the process [4,38]. In reference to compaction step, the green and sintered densities increased with pressure for compaction pressure within the range of $40\text{--}280\text{ MPa}$. Also, the variation in sintered densities for pellets produced in batches were smallest in the pressure range of $90\text{--}120\text{ MPa}$. However, in India at Bhabha Atomic Research Centre (BARC), pre-compaction step is introduced at 100 MPa to avoid chipping or breaking of the green pellets, and then final compaction of thorium dioxide can be done at higher pressures between 200 and 300 MPa [27].

4.1.3. Pellet sintering

Many additives (sintering aids) have been used and considered to have improved sinterability of pellets. This idea of using additive is to replace some of the Th^{4+} ions with metal ions that have different valency. The replacement of thorium ions will increase its non-stoichiometry, hence introduce oxygen vacancies and interstitials. The presence of these oxygen vacancies and interstitials enhance diffusion of thorium ions and consequently, more homogeneous and higher density pellets. The sintering aids such as MgO and Nb_2O_5 has been used in the past and found to reduce the sintering temperature from $1600\text{--}1700^\circ\text{C}$ to $1150\text{--}1450^\circ\text{C}$. The divalent additives tend to perform better under reducing atmosphere while niobate do better under oxidising atmosphere [27]. Sintering of thorium oxide pellets should be done in an appropriate atmosphere which is determined by the nuclide (i.e. either uranium or plutonium) in the fuel. The sintering process normally starts at temperature between 1600 and 1700°C at inert atmospheric pressure. If an ideal oxidising atmosphere for Th-MOX fuel cannot be achieved in the furnace due to safety measure, the pellets will be given second heating in the air, which will reduce the temperature between 1000 and 1200°C . Therefore, this process if properly applied will produce better oxygen stoichiometry and higher fuel density for pellets. Fuel pellets produced in this method are deemed successful if their densities are equal or greater than 96% of TD [4,38]. Furthermore, care must be taken to ensure that the fuel pellets are stored in dry conditions, as water is easily adsorbing to thorium dioxide surface. It is worthy to note that the above processes were performed without adding binder materials [27]. Fig. 9 shows the flow sheet of fabrication of $(\text{Th}, \text{U})\text{O}_2$ pellets.

4.2. Powder-plasticiser mixtures

This process of fuel fabrication uses binder materials. This process popularly called “Coated Agglomerate Pelletisation (CAP) Technique” has been developed and extensively used in India at Bhabha Atomic Research Centre. It is proposed to replace the conventional powder metallurgy method that involves direct blending of $^{233}\text{UO}_2$ and ThO_2 powders. This method is used to fabricate microsphere for TRISO or Sphere-Pac/VIPAC fuels from ThO_2 powder by adding binder materials [27]. This technique is planned to be performed in two segments, one in an unshielded facility and the other in a shielded facility. In this CAP process, freshly prepared ^{233}U oxide and ThO_2 are used to minimise man-rem problems [48]. The main objective of developing this method for the fabrication of $(\text{Th}, \text{U})\text{O}_2$ are as stated below:

1. To minimise dust in fuel fabrication operations as noticed in powder metallurgy process.
2. To decrease the number of stages followed inside shielded cell or glove box.
3. To reduce the man-rem problems since ^{233}U whose daughter isotope is a high gamma emitter would be restricted to only certain steps in the process.

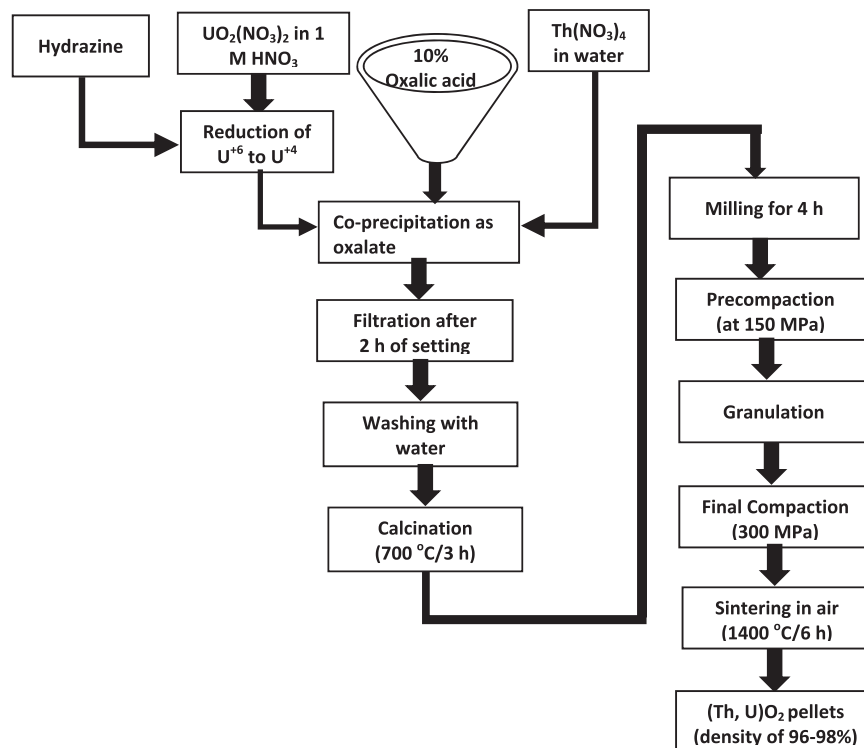


Fig. 9. Flow-sheet of fabrication of ThO₂–UO₂ pellets by co-precipitation process [72].

However, in this process, ThO₂ powder which will be in the form of flowing agglomerate is mixed with a plasticiser (e.g. paraffin, or petroleum mixture) at slightly high temperature. Afterward, the plastic mixture is moulded in an extruder and converted to agglomerate in a spheroidiser to form small spheres. The agglomerates are then dried to remove the plasticiser (organic binder). These stages are done in an unshielded facility. The next stages are done in a containment and shielded facility. In this stage, the agglomerated ThO₂ sphere will be coated with desired amount of ²³³U layer, then pressing into green pellets by compacting in a multi-station rotary, followed by the sintering of the pellets in air or reducing atmosphere, and lastly pellets loading and making them into fuel rods. The result obtained is encouraging as dust formation is minimised and the number of processes in shielded facilities reduced. Also, the results show that the agglomerate size, coating technique, compacting and sintering parameters all have great role in the homogeneity of the fissile uranium distribution in the thorium oxide matrix. The microstructure of (Th, U)O₂ fuel pellets produced showed a rock-in-sand structure with small grains in the centre of granules and large grains along the periphery as shown in Fig. 8. Indeed, more researches are still progressively being conducted at Bhabha Atomic Research Centre on Th-MOX fuels for more reliable results that would stand technical test and reactor qualifications [68].

4.2.1. Impregnation technique

This technique is still under development and investigation stage, especially at BARC. The intention and advantage of this technique is to simplify conversion steps and eliminate many expensive processes in the fuel fabrication such as precipitation of ammonia diuranate, calcination, mixing, grinding, granulation, etc., which are known to be associated with radiotoxic hazards. For this technique, low density ($\leq 75\%$ TD) ThO₂ pellets with open porosity are prepared in an unshielded facility. This is followed by sintering to produce high density ThO₂ based mixed oxide pellets. This method has been used for the fabrication of ThO₂–UO₂ pellets fuel at BARC. At BARC, the process involves fabrication of low density (about 66% TD) ThO₂ pellets by powder routes in an unshielded facility, followed by the impregnation

of the pellets using uranyl nitrate solution in vacuum in a shielded facility, and then drying of the impregnated pellets and sintering occurs at 1700 °C temperature in a reducing atmosphere. Variation of the loading of uranium in ThO₂ pellets would be done to control the density of the pre-sintered ThO₂ pellets, the concentration of uranyl nitrate solution, and the impregnation period. The density of the sintered pellets was in the range of 93–96% TD. The spider web-like microstructure in the pellets is shown in Fig. 11. This is different from the sand-like granular microstructure observed in Fig. 10 for ThO₂–UO₂ fuel using CAP method [68]. As hinted earlier, this method is still under investigation for possible industrial application.

4.3. Sol-gel methods

This method is developed as an alternative to the conventional powder mixing fuel fabrication technology. The sol-gel method is judged to be mostly suitable for (Th, ²³³U) MOX fuel manufacture. This process uses liquids or free flowing spherical solids which are easily adaptable for automated and remote operations [27,68]. There are two major sol-gel routes used for fuel fabrication; internal gelation and

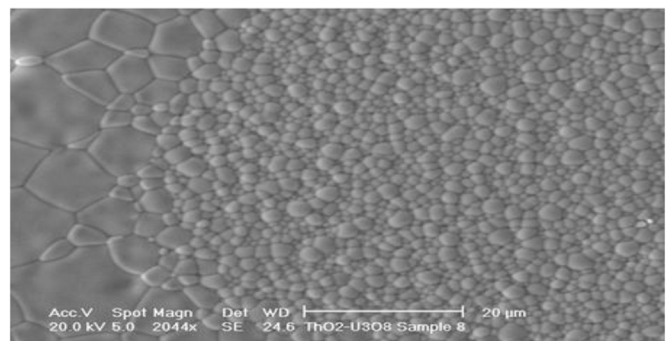


Fig. 10. Microstructure of ThO₂–4%UO₂ pellet made by CAP process showing duplex grain structure. Pellet sintered in air and etched thermally [68].

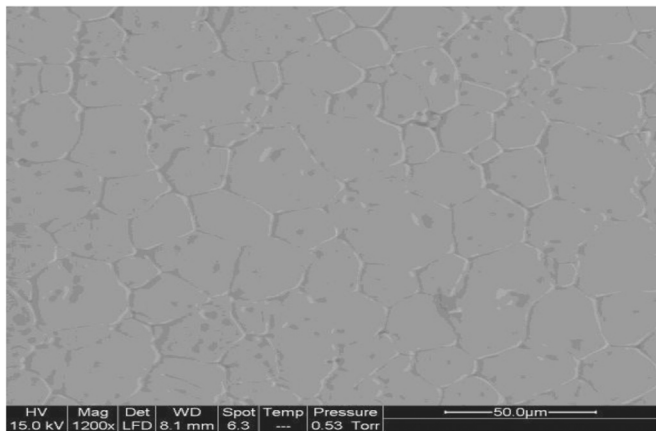


Fig. 11. The spider web-like microstructure of ThO_2 -4% UO_2 pellet made by impregnation [68].

external gelation of thorium. The internal gelation of thorium process is developed in Netherlands for uranium and plutonium but has been used successfully for thorium in India and Germany while the external gelation process was developed in Germany and the United States at ORNL. Generally, sol-gel technique begins with the nitrate feed solution as in powder compaction but does not use oxalic acid rather ammonia is added, which sparks off the formation of microcrystals and subsequent gelation of the sol [73]. The resulting sol-gel structure is washed and dried to produce sol-gel microspheres, followed by calcination and sintering of the sol-gel microsphere into pellets or Sphere-Pac column or coated to produce TRISO fuel. However, for internal gelation of thorium process, the nitrate feed solution is mixed with ammonia producing additive such as hexamethylene tetraamine and urea solutions of similar concentration at 0°C temperature while no ammonia producing additive is used for external gelation process rather the sol drops pass through gaseous ammonia which then gels the surface of the droplets [27]. At BARC, the sol-gel process uses free flowing dust-free sol-gel microspheres as a starting material instead of the powder for pellets fabrication. For quality assurance of the pellets to be achieved, the microspheres should possess low crushing strength, reasonable density ($\geq 96\%$ TD) and high surface area [68].

5. Reprocessing of nuclear fuel

Generally, to reprocess spent fuel, it is compulsory to get vital information on the inventories of the spent fuel, determining the type of fuel used, the number of years in the reactor, and the number of years it has stayed as spent fuel. These inventories help to know the type of radionuclides in spent fuel whether in open or closed cycle, or uranium or thorium-based fuels [37]. Consequently, many reprocessing methods have been developed and used both in the past and present either at the research stage or commercial scale for uranium and thorium-based reactor spent fuels. The methods are as listed below:

1. Pyrometallurgy – which uses heat to start the separation process of metals from their mineral.
2. Electrometallurgy – this uses electric current to initiate the separation process of the metals. It is also called pyroprocessing and has a good prospect though not yet commercially operational.
3. Hydrometallurgy – this employs aqueous solution in dissolving the metal content. This is the main reprocessing method that has proved efficient and reliable for reprocessing UOX or its MOX spent fuel. It is popularly known as PUREX (Plutonium Uranium Redox EXtraction), which is fully developed and commercially operational. The PUREX reprocessing flow sheet is shown in Fig. 12.

5.1. Reprocessing of spent thorium fuel

Thorium-based fuel as an emerging and promising nuclear fuel would be used in reactor mostly on closed cycle, and the fissile ^{233}U bred from thorium is the needed fuel. Therefore, thorium-based fuel reactors must incorporate adequate reprocessing technology to its design configurations.

The process for reprocessing spent fuel from thorium fuel reactors is developed based on PUREX technology. This process of reprocessing spent thorium fuel is popularly called THOREX (THORIum-uranium EXtraction), which is based on solvent extraction of thorium and uranium by TBP that was developed at ORNL in the middle of 1950s and is judged the most viable route for recovering uranium and thorium [27]. The gain in the stability properties of thorium in the reactor operation is now a challenge making ThO_2 and $(\text{Th}, \text{U})\text{O}_2$ or $(\text{Th}, \text{Pu})\text{O}_2$ fuels insoluble in nitric acid solution used to dissolve the spent fuel in the PUREX process. Hence, to dissolve spent thorium-based fuel, the THOREX reagent [$13\text{ M HNO}_3 + 0.05\text{ M HF} + 0.1\text{ M Al}(\text{NO}_3)_3$], which is a mixture of nitric acid, fluoride ion and aluminium nitrate is used. The fluoride ion is added to the nitric acid to catalyse the dissolution reaction while aluminium nitrate helps to deactivate the corrosion effect of fluoride on the stainless-steel equipment in the reactor. Fig. 13 shows the THOREX flow sheet for the reprocessing of spent thorium-based fuels. However, THOREX process has not attracted commercial industrial application as it is still used in the laboratories and pilot plant scale [38]. THOREX process has the following challenges that must be addressed:

1. From the reaction chain of thorium-232, protactinium-233 (Pa) exist as an intermediate product with a half-life of 27 days longer than its counterpart neptunium-239 (Np) in uranium cycle that has a half-life of 2.4 days. Because of this, and to avoid loss of ^{233}U , a decay period of about 12 months should be allowed before carrying out reprocessing to ensure that all the ^{233}Pa in the spent fuel has decayed completely to ^{233}U . Also, protactinium could be found in the extracted fission product waste, and this can have radiological issue since ^{231}Pa , an alpha emitter could be formed in the decay process. Presently, novel method [74] can be used to remove ^{233}Pa from the fission product waste thereby solving the problem.
2. The decay chain of thorium-232 will also produce ^{232}U which is an alpha emitter, and then decays to produce daughter elements that emit high energy gamma radiations. Typically, ^{232}U decays to ^{208}Tl which is the deadliest among its daughter elements emitting 2.6 MeV gamma radiation. The production of ^{232}U is smaller in thermal reactors but higher in fast reactors containing thorium-based fuels. Although, ^{232}U is not a problem if it remains in the reactor, because it will easily produce ^{233}U by neutron capture reaction [31,32]. Indeed, these issues surrounding THOREX process of reprocessing spent fuel require that more reprocessing test and research should be conducted before its use for commercial application. And, without closed cycle operation for thorium-based fuel, it will limit the intended gain on uranium resource savings.

6. Prospects and drawbacks of thorium fuel cycle

6.1. Advantages of thorium-based nuclear fuel

The driving prospects of thorium-based nuclear fuels in reactor operation are summarised as follows:

1. Thorium is known to be three to four times higher in abundance than uranium, thereby has the potential of ensuring nuclear fuel sustainability for the world increasing energy need [6,75].
2. The characteristic physical and chemical stability of thorium and its compounds (especially oxides) are of great importance. These stability properties enhanced its potential use in MOX fuel (that is as

PUREX flow sheet for reprocessing used fuel

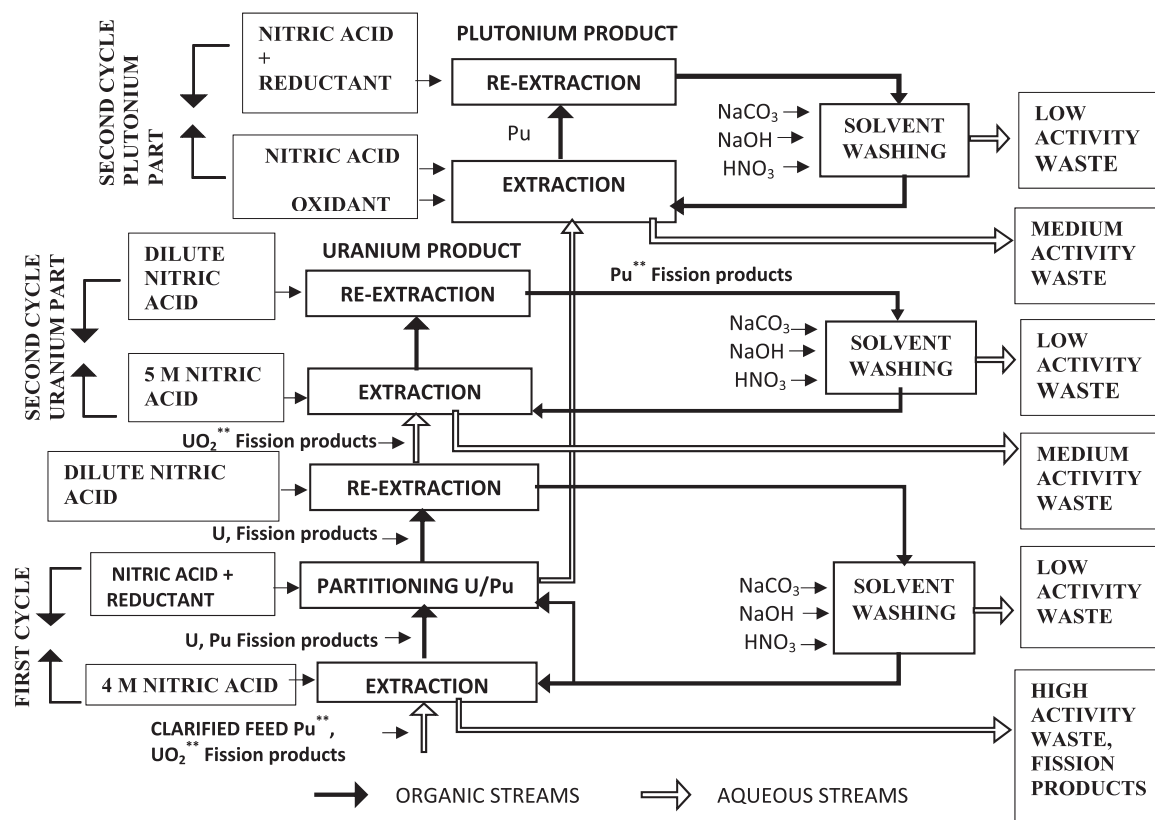


Fig. 12. The Plutonium Uranium Redox Extraction (PUREX) flow sheet [38].

[(Th, ^{233}U), (Th, U) O_2 , (Th, Pu) O_2]], which would reduce drastically the large stockpiles of both reactor and weapon-grade plutonium from uranium-based conventional reactors. These properties of thorium will also minimise the proliferation of nuclear materials and global terrorism concerns.

3. The production rate of transuranic elements by ^{232}Th fertile isotope is much lower compared to fertile ^{238}U , as about four successive stages are required to produce a transuranic while only one stage is required in the case of ^{238}U . Also, the production of long-lived minor radioactive actinides resulting from fission by thorium-based fuels are much lower in quantity compared with uranium or plutonium-based fuel cycles as shown in Table 6. These positions would generally minimise the challenges of managing spent fuels and nuclear waste.
4. ^{233}U bred from thorium, has a higher average number of neutrons produced per neutron capture compared to ^{239}Pu and ^{241}Pu in (U, Pu) cycle, resulting to a higher neutron yield for thorium fuel cycle. Although, it has lower capture per fission reaction rate compared to ^{239}Pu .
5. ThO_2 has higher melting temperature of 3000–3300 °C compared to UO_2 that melts at around 2700–2800 °C. This thermal stability gives room for high burn-up of thorium fuels in reactors.
6. ^{233}U maintains its good neutronic properties with higher temperature better than that of ^{235}U and ^{239}Pu . This gave thorium fuel cycle the potential for use in high temperature reactors.
7. The thorium fuel cost will be reduced up to 20–30%, especially in the reactor design configuration that requires little or no uranium enrichment.
8. The cost of building and operating thorium-based fuel reactors would reduce, since thorium-based fuel can be used in the conventional reactors such like LWRs, CANDU, HTR, etc. with minimal

technological adjustment. Also, many of the thorium fuel cycle reactors will possibly operate on closed cycle system.

9. Most importantly, there is a feasibility of achieving molten salt reactor, fast breeder reactor, Accelerator Driven System reactor in the nearest future, which are more politically acceptable based on their inherent properties.

6.2. Drawbacks of thorium-based nuclear fuel

Despite the significant progress that has been made at different stages for possible thorium fuel application at commercial level in nuclear reactors, there are needs for more experimental research programmes to address some technological uncertain areas before a commercial industrial application can be achieved. Some of these areas are as follows:

1. The initial operating fissile material requirement for thorium fertilisation is a major concern in thorium fuel cycle application. This initial reactor start-up fissile isotope will still generate long-lived actinides and transuranic elements, as most of the thorium-based reactor design configurations requires HEU or low but very near to HEU, which also, does not fall within the 3–5% IAEA recommended for uranium enrichment. Though, the quantities of transuranic and long-lived actinides would be much smaller compared with that of conventional reactors in operation, it is still a concern in most proposed thorium fuel reactors. To find solution to this for thermal reactors, research is ongoing using the Radkowsky seed-blanket core model. Moreover, this issue can be removed in the MSRs and ADSRs reactor design configurations, which have the capacity to burn the problematic long-lived actinides and transuranic elements during reactor operations [38].

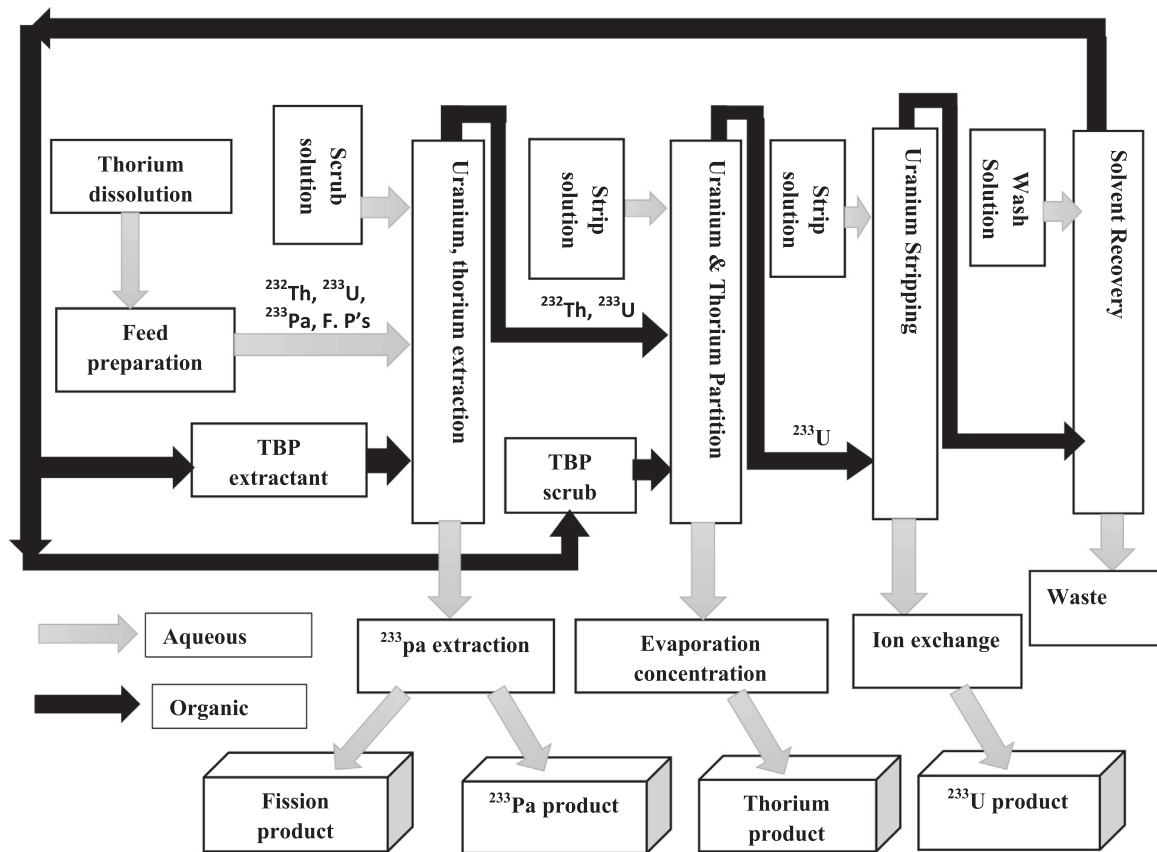


Fig. 13. The THOREX flow sheet for the reprocessing of spent thorium-based fuels [38].

Table 6
Relative production of minor actinides in different fuel cycles [33].

Minor actinides	Production from different fuel cycles			
	$\text{U}^{235} + \text{U}^{238}$ (Reference cycle g/t)	$\text{U}^{235} + \text{Th}^{232}$ (%)	$\text{U}^{233} + \text{U}^{238}$ (%)	$\text{U}^{233} + \text{Th}^{232}$ (%)
Np-237	360 [*] 900 ^{**}	92 107	20 13	1 3
Am (241 + 242 + 243)	160 [*] 470 ^{**}	0.04 0.28	106 117	6.3×10^{-5} 1.8×10^{-3}
Cm (243–246)	36 220	0.01 0.14	111 132	1.6×10^{-5} 6.37×10^{-4}

* Discharge burn-up of 30GWd/t, and.

** Discharge burn-up of 60GWd/t.

- To breed ^{233}U from ^{232}Th takes relatively a long interval due to the 27 days half-life of ^{233}Pa as the reaction intermediate isotope. This means that a reasonable amount of ^{233}Pa , which is a good neutron absorber will build-up in thorium-based fuel. Although, ^{233}U will later produce another fissile ^{235}U isotope, it requires two more neutrons absorption to do that reducing the initial number of neutrons for fission and increases the chance of more transuranic elements in spent fuel. This makes it difficult to design thorium fuel cycle that produces more ^{233}U in thermal reactors than the start-up fissile isotope it consumes, even though ^{233}U in a reactor has high fuel conversion ratio and longer fuel burn-ups than either ^{235}U or ^{239}Pu . The effect of ^{233}Pa is not a major problem as it is estimated to be 1/1000 by weight in thorium fuel, and some reactor designs such as MSRs, has a breeding ratio (BR) of 1.07 with online removal of ^{233}Pa and 1.05 without the ^{233}Pa removal [29].
- The stability and single oxidation state of thorium makes chemical treatment for the separation of thorium compounds from spent fuel

very difficult, and reprocessing is a must in reactor designs to extract ^{233}U from the spent fuel for use as new fuel. It is easier to separate and process the waste in spent fuel from the conventional uranium-based fuel cycles than in thorium-based reactors.

- The characteristics of thorium made it a good candidate for MOX fuel. The implication is that reprocessing will be an integral part of most thorium-based reactors, which is not too easy as stated above, except for few reactor designs.
- The dissolution of thorium dioxide is very difficult compared to uranium dioxide. But this issue can be solved by adding buffered fluoride in the dissolvent, and careful addition of complexing agent such as aluminium nitrate to prevent anticipated corrosion of stainless steel equipment, which would have occurred due to buffered fluoride. This issue is still a problem in LWRs with zircalloy cladding, as the zircalloy will dissolve in the buffered fluoride complicating reprocessing of spent fuel, thereby increasing the production of krypton more than ^{233}U [33].
- The qualification and certification of thorium nuclear data for reactor designs, safety analysis, reactor operation, transport and storage need further evaluation and test to avoid the existing type of problems that surrounds uranium-based fuel reactors. Also, on March 2011, Fukushima Daiichi nuclear accident in Japan [15,16], that was caused by earthquake though was successfully managed based on the reactors safety design, but the subsequent tsunami caused the explosion [13], which was not incorporated in the earlier reactor safety designs. This has further stressed the need for proper qualification of all processes before thorium reactor operation at commercial level.

7. Thorium fuel research trend and development

In this section, thorium research development and trend that are

hoped to provide solution to the drawbacks discussed in Section 6.2 and the summary of the impact of the research are presented. The drawbacks indeed do not seem insurmountable as there are many researches being carried out by various research institutions and agencies. Such proposals with nearer-future application are; the Radkowsky seed-blanket core model, which has been used in Russian water-water reactor (popularly known as VVER reactors) design as discussed in Section 3.1. and the pebble-bed or prismatic core very high temperature reactor (VHTR) design discussed in Section 3.3.1, tipped by GIF for hydrogen and electricity production. Other researches are being conducted on the molten salt reactor, Fast breeder reactor, and accelerator driven reactor designs for far-future applications.

7.1. Radkowsky thorium reactor (RTR) concept

The concept of RTR was introduced to enhance the use of thorium fuel cycle in all reactor types, especially light and heavy water reactors. This concept has a seed-blanket reactor core design with the uranium fuel seed and thorium fuel blanket, thereby mitigating the challenges of using fertile natural thorium fuel which requires initial fissile nuclide introduction for thermal fission reactions to occur. The novelty of this is also on its in-core fuel management, which does not require reprocessing of spent fuel [55,76,77]. The viability of this concept was investigated in a collaborative arrangement by the Radkowsky Thorium Power Corporation and Kurchatov Institute in Russia, with technical contributions from the Brookhaven National Laboratory (BNL) [76]. RTR design offers higher proliferation resistance, higher fuel burn-up, and less challenging waste because of the favourable inherent characteristic properties of thorium. It is worth mentioning that this concept has been successfully used in Russian light water reactors (VVERs) as reported by Galahom [6], Galperin et al. [41], Lee and Cho [77], and more studies are being conducted for its application in Advanced Heavy Water Reactors (AHWR) in India and Canada [18,55,56]. The use of thorium mixed oxide fuel is also being tested in existing light water reactors especially PWR and PHWR designs based on the high burn-up properties of thorium fuel and reduced plutonium production [78,79]. Summarily, this thorium reactor core design as used in VVER, Indian PHWR and AHWR has therefore, proved successful and provided solution to the challenges of initial fissile fertilisation, reduced waste management issues, proliferation risk, and providing hope for its successful use in most LWRs, as thorium-based fuel is currently being used for power peaking control in PWR design [56]. Currently, Massachusetts Institute of Technology (MIT) and BNL are collaborating on further studies to underscore the thermal and neutronic properties of seed-blanket reactor core for different lattice (triangular and square) arrangements [55].

7.2. Thorium research at CERN and other nuclear centres

As discussed in Section 3 on the use of thorium fuel in different reactor types, several research groups are investigating the long-term applications of natural thorium fuel without fissile nuclide at European Organisation for Nuclear Research (CERN), Los Alamos National Laboratory (LANL) etc. for molten salt reactors, fast reactors and accelerator driven system reactors. These studies for molten salt thermal energy amplifier and fast energy amplifiers are based on the conceptual design proposed by C. D. Bowman at LANL and Carlos Rubbia at CERN respectively. Today, based on Generation-IV International Forum (GIF) recommendations, other nuclear research institutions under the watch and regulation of IAEA are collaborating in this research. Presently, there is high hope of achieving success based on the IAEA Technical Working Material report [56,80]. There are still wide areas to cover in thorium research before further evaluation and qualification of thorium fuel cycle for possible commercial application.

8. Conclusion

This review successfully investigated the prospects and drawbacks of thorium-based fuel cycle. Many theoretical and experimental researches have been conducted on thorium as a possible nuclear fuel spanning from the late 1940s till date. The nuclear technology leaders such as USA, Germany, Canada, France, Japan, United Kingdom, and other countries like India, China and Russia have made significant contributions in nuclear industry, and particularly towards thorium based nuclear fuel research programmes. Many dedicated researches on thorium fuel cycle have been conducted at ORNL, MIT, BARC, LANL, Battelle Memorial Institute, Westinghouse Electric Company and so many other laboratories, agencies and institutions, with encouraging results. At the earlier stage of nuclear technology, several researches were conducted on thorium with the intention of producing weapon grade material. This was not the case, hence the low interest in thorium seen between 1950s and early 1970s. But late 1970s and 1980s saw rapid increase in thorium researches, with many experimental and prototype test reactors designed and operated. The issues of uranium abundance, weapon-grade plutonium, radiotoxicity, waste management and nuclear terrorism were the factors that made thorium fuel cycle to stand out more than uranium-based fuel reactors. The stable characteristics (physical, thermochemical and neutronic properties) of thorium and its bred ^{233}U made thorium a better nuclear material over uranium. These properties of thorium and its bred have been studied by many researchers, agencies and organisations. Thorium alone is difficult to be used as fuel in a thermal reactor because it requires a fissile fertilisation nuclide. But this is not a problem since ThO_2 fuel has high potential to be used as MOX fuel with fissile plutonium and uranium oxides, which will minimise the large stockpiles of plutonium. However, thorium-based fuel proved successful under irradiation in high temperature gas cooled reactors, light water reactors, heavy water reactors, molten salt reactors, and would be the best in accelerator driven system reactors.

Based on this review and for near-term application, thorium-based fuel can be used in all existing LWRs designs with little technical modifications to the initial reactor design configurations. Indeed, this gave a strong hope for its near-term commercial applications in LWRs, HWRs, HTGRs and MSRs, while ADSRs and fast breeder reactors, and should be considered for long-term application. The earlier envisaged economic lost if the present LWRs reactors were to be dismantled in favour of new thorium fuel reactor designs is already minimised by the reported potential of thorium fuel efficiency in existing reactors with minimum modification [5,8].

Thorium data have been collated, and technical parameters analysed by world nuclear bodies such as IAEA, NEA, OECD and GIF, for structural, material and engineering qualification. In lieu of this, Generation IV International Forum (GIF) has a projection of 2030 time frame for possible commercial thorium-based fuel reactor, which most likely will be for LWRs, HTGRs and molten salt reactor designs. Therefore, more experiments and test researches for thorium-based fuel reactor configurations should be conducted for both short- and long-term applications to ensure that its structural, engineering and irradiation behaviours are adequately qualified. Finally, thorium-based nuclear fuel cycle has the potentials of providing sustainable fuel, higher fuel burn-up, longer reactor life cycle, less problematic waste, and more politically acceptable and environmentally friendly clean energy compared to the conventional uranium-based fuel cycle.

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